

Name :- Mihir Mungara

Airflow Assignment – 5

Assignment Definition :- Create DAG for demo dynamic task pipeline so first task will generate 10 random of number between 1 to 100 and it will pass list of it to next task through Xcom and using dynamic task there will be a task which will categories number into 3 segments based on number's value (Segment : Great, Average, Poor) and Last task will display number with segment value.

Code :-

```
from airflow import DAG
from airflow.decorators import task
from datetime import datetime
import random

with DAG(
    dag_id="Assignment-5",
    start_date=datetime(2026, 1, 1),
    schedule=None,
    catchup=False,
    tags=["dynamic-task-mapping"]
) as dag:

    # =====
    # TASK 1: Generate 10 random numbers
    # =====
    @task
    def generate_numbers():

        numbers = [
            random.randint(1, 100)
            for _ in range(10)
        ]

        print(f"Generated numbers: {numbers}")

        return numbers

    # =====
    # TASK 2: Dynamic Task Mapping
    # =====
```

```

@task
def categorize_number(number):

    if number >= 70:
        segment = "Great"

    elif number >= 40:
        segment = "Average"

    else:
        segment = "Poor"

    result = {
        "number": number,
        "segment": segment
    }

    print(result)

    return result

# =====
# TASK 3: Display Results
# =====
@task
def display_results(results):

    print("\nFinal Categorization:\n")

    for result in results:

        print(
            f"Number: {result['number']} "
            f"→ Segment: {result['segment']}"
        )

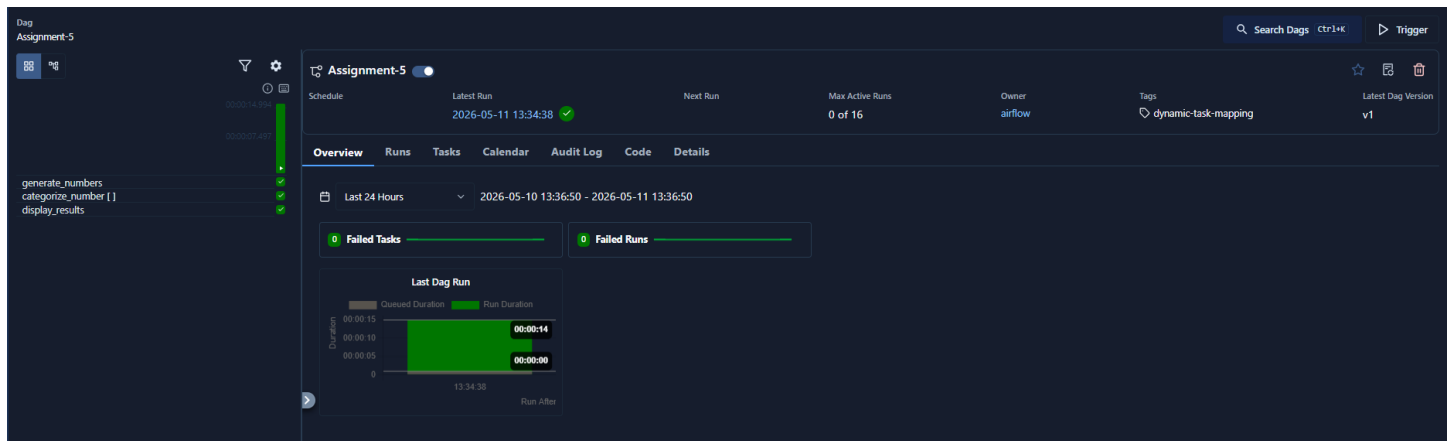
# =====
# DAG FLOW
# =====
random_numbers = generate_numbers()

categorized = categorize_number.expand(
    number=random_numbers
)

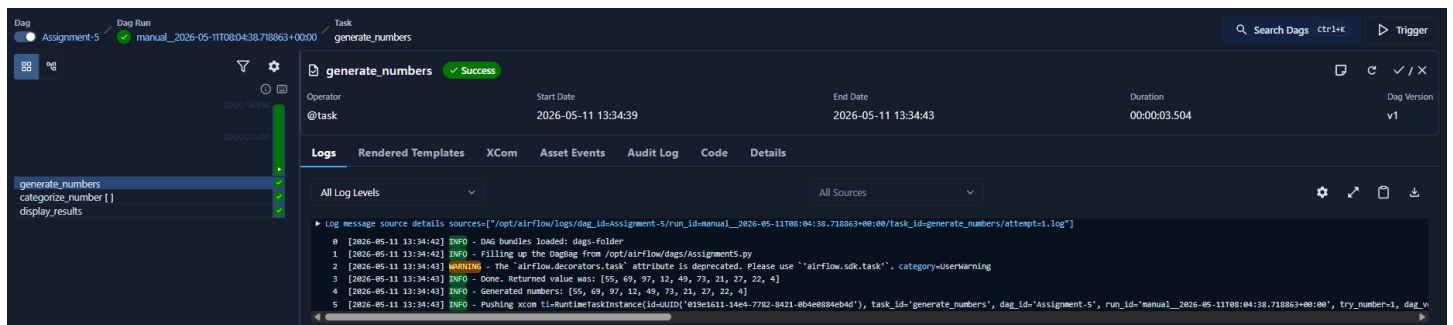
display_results(categorized)

```

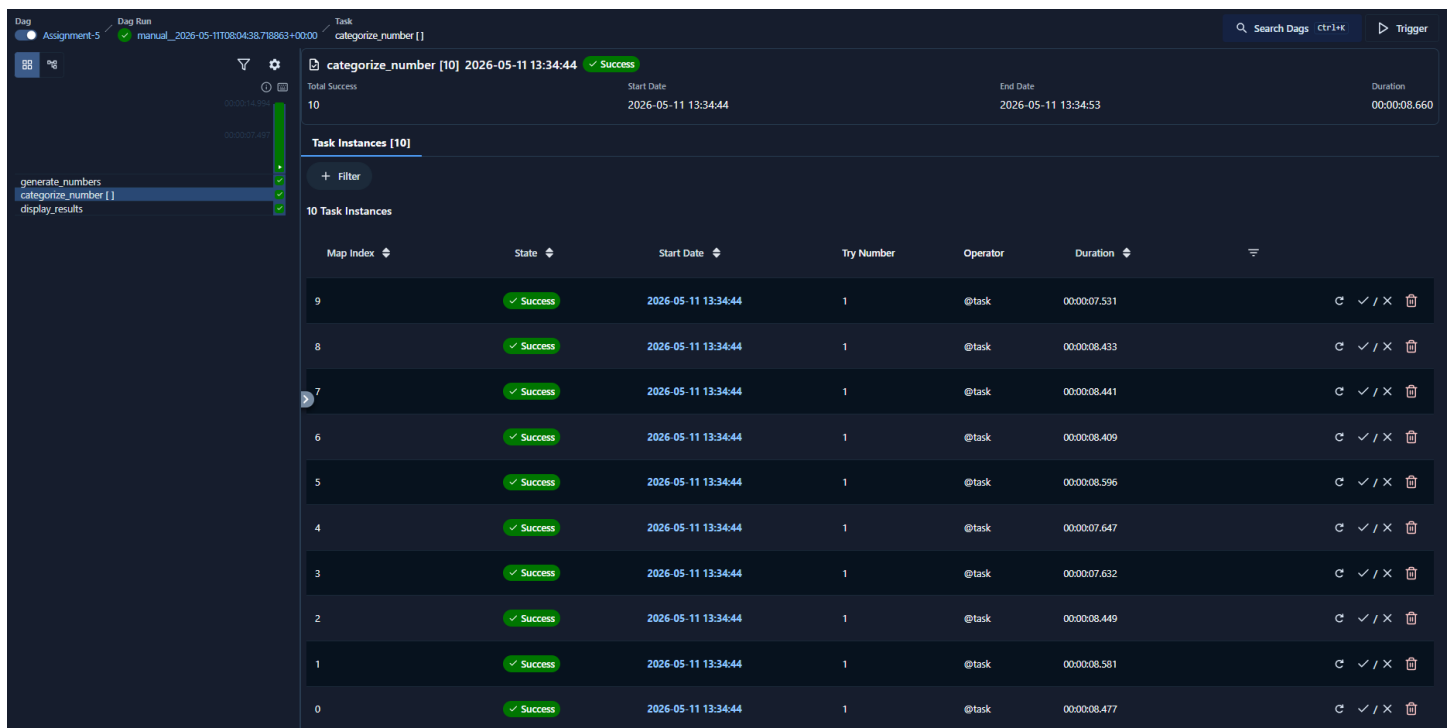
Airflow DAG Home Page :-



Showing Logs of Generating Numbers Task:-



Showing Logs of Categorizing Numbers Task :-



Showing Logs of Any one of the 10 Categorized tasks :-

The screenshot displays the Airflow web interface. On the left, a task list shows 'generate_numbers', 'categorize_number []', and 'display_results'. The 'categorize_number []' task is selected. The main panel shows the task details for 'categorize_number [0]' with a status of 'Success'. The task was executed by '@task' on 2026-05-11 13:34:44, with a duration of 00:00:08.477. The 'Logs' tab is active, showing log messages from the task's source file. The logs include information about DAG bundles, task filling, and the final categorization of the number 55 as 'Average'.

Operator	Map Index	Start Date	End Date	Duration	Dag Version
@task	0	2026-05-11 13:34:44	2026-05-11 13:34:53	00:00:08.477	v1

Logs Task Instances [10] Rendered Templates XCom Asset Events Audit Log Code Details

All Log Levels All Sources

```
> Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-5/run_id=manual_2026-05-11T08:04:38.718863+00:00/task_id=categorize_number/map_index=0/attempt=1.log"]
0 [2026-05-11 13:34:45] INFO - DAG bundles loaded: dags-folder
1 [2026-05-11 13:34:45] INFO - Filling up the Dagbag from /opt/airflow/dags/Assignments.py
2 [2026-05-11 13:34:51] WARNING - The 'airflow.decorators.task' attribute is deprecated. Please use 'airflow.sdk.task'. category=UserWarning
3 [2026-05-11 13:34:53] INFO - ('number': 55, 'segment': 'Average')
4 [2026-05-11 13:34:53] INFO - Done. Returned value was: ('number': 55, 'segment': 'Average')
5 [2026-05-11 13:34:53] INFO - Pushing xcom ti=hantimeTaskInstance(id=AUUD('019e1611-14e5-7dd8-baca-7af6b3af267e'), task_id='categorize_number', dag_id='Assignment-5', run_id='manual_2026-05-11T08:04:38.718863+00:00', try_number=1, dag='Assignment-5')
```

Displaying Result Task :-

The screenshot displays the Airflow web interface. On the left, the task list shows 'generate_numbers', 'categorize_number []', and 'display_results'. The 'display_results' task is selected. The main panel shows the task details for 'display_results' with a status of 'Success'. The task was executed by '@task' on 2026-05-11 13:34:54, with a duration of 00:00:00.387. The 'Logs' tab is active, showing log messages from the task's source file. The logs include information about DAG bundles, task filling, and the final categorization of the number 55 as 'Average'.

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-05-11 13:34:54	2026-05-11 13:34:54	00:00:00.387	v1

Logs Rendered Templates XCom Asset Events Audit Log Code Details

All Log Levels All Sources

```
> Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-5/run_id=manual_2026-05-11T08:04:38.718863+00:00/task_id=display_results/attempt=1.log"]
0 [2026-05-11 13:34:54] INFO - DAG bundles loaded: dags-folder
1 [2026-05-11 13:34:54] INFO - Filling up the Dagbag from /opt/airflow/dags/Assignments.py
2 [2026-05-11 13:34:54] WARNING - The 'airflow.decorators.task' attribute is deprecated. Please use 'airflow.sdk.task'. category=UserWarning
3 [2026-05-11 13:34:54] INFO -
4 [2026-05-11 13:34:54] INFO - Final Categorization:
5 [2026-05-11 13:34:54] INFO -
6 [2026-05-11 13:34:54] INFO - Number: 55 -> Segment: Average
7 [2026-05-11 13:34:54] INFO - Number: 69 -> Segment: Average
8 [2026-05-11 13:34:54] INFO - Number: 97 -> Segment: Great
9 [2026-05-11 13:34:54] INFO - Number: 12 -> Segment: Poor
10 [2026-05-11 13:34:54] INFO - Number: 49 -> Segment: Average
11 [2026-05-11 13:34:54] INFO - Number: 73 -> Segment: Great
12 [2026-05-11 13:34:54] INFO - Number: 21 -> Segment: Poor
13 [2026-05-11 13:34:54] INFO - Number: 27 -> Segment: Poor
14 [2026-05-11 13:34:54] INFO - Number: 22 -> Segment: Poor
15 [2026-05-11 13:34:54] INFO - Number: 4 -> Segment: Poor
16 [2026-05-11 13:34:54] INFO - Done. Returned value was: None
```